Off-centre CoE cosmology: a toy universe from geometry to the Planck TT spectrum, on one consistent background

Working notes for C. A. (revision 7, compiled 25 October 2026)

Contents

1 Introduction **2**

2 The model **5**
2.1 The question and set-up . . . 5
2.2 One background: the LTB famP solution . . . 5

3 Methods **7**
3.1 The inverse approach . . . 7
3.2 Spherical harmonics and D_ℓ . . . 8
3.3 Goodness of fit . . . 8
3.4 Model comparison and parameter accounting . . . 9

4 Data **9**

5 Results **9**
5.1 Density laws, the CatWISE lock and the Migkas fit . . . 9
5.2 Correlated bins and model comparison for the cluster profile . . . 11
5.3 The CMB: the deterministic sky at last scattering . . . 12
5.4 Compensating structure inside the last-scattering sphere . . . 12
5.5 Stochastic field and sky projection . . . 13
5.6 TT fit . . . 14
5.7 Distribution of skies . . . 14
5.8 Joint summary . . . 15
5.9 Open CMB problems . . . 15
5.10 Sky anomalies . . . 16

6 Discussion **16**

7 Conclusions **18**

References **19**

A Parameter accounting **20**

B Figure provenance **20**

C Changes from rev6 **21**

D Response to referee, audit and internal review **21**

Reading this draft

This revision rewrites rev6 around one background (an LTB solution with the observer at $D = 34.96\,\mathrm{Gpc}$) and answers the referee report, the mathematics-and-code audit, and the internal five-pass review point by point (Appendix D). Every figure was regenerated from the current solve or is new, and each PNG has its generating `.py` beside it in `fig7/`. Figures and analyses that are new in this revision are tagged [**NEW in rev7**] in their captions and in Appendix B. Rev6 figures that could not be regenerated from the current solve are not carried over (Appendix B lists them). [TO ADD: …] marks a document or date that I do not have. [TO COMPUTE: …] marks a number that still needs a run. I have not invented a value for either. [unverified: …] keeps its earlier meaning: the item is not traced to code or data. The framing throughout is that of a *toy universe*: the aim is to show, by numerical solution, that the effects go in the right direction. It is not a precision fit. The inverse problem is mentioned, not developed.

Abstract

We describe a toy universe in which a finite expansion of ordinary matter has a centre of expansion (CoE), and the observer sits off that centre, inside a thick shell. Because the symmetry is axial about the CoE axis rather than spherical about the observer, the sky has a preferred direction. In this revision all parts of the calculation use one background: a Lemaître–Tolman–Bondi (LTB) solution with the observer at $D = 34.96\,\mathrm{Gpc}$ from the CoE. In this solution the local expansion rate is $H_0 = 34.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, the age on the observer's shell is 0.314 Gyr, and the comoving radius of the last-scattering sphere is $R_{\rm ls} = 0.190\,\mathrm{Gpc}$. These are far from the standard values, and we state them as a tension of the toy model, not as estimates.

Predictions made before the data. An earlier point-mass Newtonian version of the model predicted the two-hump shape of the cluster normalisation curve $A/A_{\rm all}$ against angle, ring positions on the cluster sky, and the sign of a CoE→anti-CoE density gradient, before Migkas et al. (2020) and Secrest et al. (2021) published. [TO ADD: dates and documents for each prediction.]

(i) Clusters. Against the 71 overlapping longitude cones of Migkas et al., the CatWISE-locked CoE profile gives $\chi^2 = 9.25$ (isotropy: 45.14) if the bins are treated as independent. With an approximate overlap covariance (about 6 effective bins), it gives $\chi^2 = 3.88$ against 8.99 for isotropy, 6.90 for a free dipole and 5.20 for a dipole plus quadrupole. The ordering favours the CoE geometry, and its BIC is lowest. None of the differences is significant ($p \approx 0.3$–0.6), so we describe the cluster result as qualitative.

(ii) CatWISE. The CatWISE dipole fixes the local density gradient through a lock whose assumptions are listed in §5.1.

(iii) CMB. Carried to last scattering at $R_{\rm ls}$, the deterministic CoE sky has rms 2.3×10^{-5}. A set of 975 clusters and 904 voids inside the last-scattering sphere, none at the mass bound, cancels the dipole and octupole to $< 10^{-9}$ and reduces D_2 to the target to within 1.5×10^{-6} in $\Delta T/T$, while leaving the cluster fit unchanged ($\Delta\chi^2 = 2 \times 10^{-7}$). The stochastic tidal field, with a 15-parameter phenomenological source spectrum, gives $\chi^2_{2-2500} = 2{,}616$ for the ensemble mean, against 2,558 for ΛCDM ($\Delta\mathrm{BIC} = +128$). Like for like, over 1,000 realisations, the median one-sky χ^2 is 3,347 for the model and 3,279 for ΛCDM. By standard statistics ΛCDM still fits better, and the acoustic structure enters through the fitted source spectrum. It is not derived.

1 Introduction

The standard cosmological model rests on the cosmological principle: on large scales the universe is homogeneous and isotropic, and every observer sees the same expansion. The inferred acceleration of that expansion, assigned to dark energy or Λ, comes from supernova distances that lie above the prediction of a decelerating homogeneous model. Several observations sit uneasily with exact isotropy:

- The X-ray scaling relations of galaxy clusters vary with direction (Migkas et al. 2020, 2021).
- The number-count dipole of CatWISE quasars is about twice the kinematic dipole expected from our motion relative to the CMB (Secrest et al. 2021).

These notes ask whether an observer in an inhomogeneous expansion, off its centre, would see such anisotropies, and whether such an observer could read an apparent acceleration from distances without any acceleration taking place. The second question is posed, not answered.

What kind of paper this is. This is a *toy universe solved numerically*. The claim tested is directional: do the effects of an off-centre CoE go the right way, in the right places, with amplitudes that can be made consistent with the data on one background? It is not a parameter-estimation paper, and the goodness-of-fit numbers below are reported to show progress, not to claim a better fit than ΛCDM.

History and predictive status. This programme began over a decade ago with Dr. Rindler. Its question was whether accelerated expansion can be explained without dark matter or dark energy, using well-known physics.

1. *Point-mass model.* A first model, a point mass at the CoE with Newtonian dynamics solved numerically, produced the axisymmetric $A/A_{\rm all}$-vs-angle curve directly from the off-centre geometry. It predicted the characteristic curve and the location of rings on the two-dimensional sky.

2. *Confirmation by Migkas et al.* When Migkas et al. (2020) published $A/A_{\rm all}$ against longitude and their sky map, both features were present.

3. *B and the gradient.* The same numerical model produced $B/B_{\rm all}$ against angle, and it favoured a density gradient from the CoE toward the anti-CoE. That is the sense of the CatWISE number-count dipole published afterwards (Secrest et al. 2021).

4. *CatWISE lock.* Only then was the observed CatWISE gradient imposed as a constraint (§5.1). At the same time, the central density was allowed to rise before falling.

Table 1 separates what was predicted from what was fitted afterwards. Predictions are not subject to the look-elsewhere and parameter-count corrections of §3.4; post-hoc fits are.

The CMB as an inverse problem. The geometry that produces the cluster sky also leaves an anisotropic CMB. Two questions follow. Can the distribution of matter around the observer change $\Delta T/T$ versus angle, through tidal forces alone, so that the CMB falls within the required limit? And is there some combination of simple masses and stochastic clusters and voids between recombination and us that yields the observed spectrum? The general inverse problem (recover the matter distribution from the sky) is not attempted here. Because C_ℓ discards phases, a flexible enough inverse problem always has a solution, so we only show that *one* physically allowed configuration exists on the chosen background (§5.4), and we report the freedom it uses.

Path dependence. Structure is expected to self-assemble in chaotic, path-dependent patterns, so re-running the universe need not give the same CMB. The model therefore predicts a distribution of skies, of which ours is one draw. Section 5.7 gives that distribution for the current field (1,000 realisations) and compares it with the same construction for ΛCDM.

Table 1: Predicted versus post-hoc quantities.

Quantity	Status	Fixed by	Date / document
shape of $A/A_{\rm all}(\psi)$ (two humps, two troughs)	predicted	point-mass geometry	[TO ADD]
ring locations on the cluster sky	predicted	point-mass geometry	[TO ADD]
$B/B_{\rm all}(\psi)$ behaviour	predicted	numerical model	[TO ADD]
sign of the CoE→anti-CoE density gradient	predicted	numerical model	[TO ADD]
gradient amplitude	fitted	CatWISE lock (§5.1)	–
law shape (η, m_c), l_0	fitted	Migkas profile	–
$D = 34.96$ Gpc	fitted	Migkas D profile (Fig. 2)	–
compensating bodies (975 + 904)	solved for	CMB low-ℓ isotropy	–
15 source-spectrum parameters	fitted	Planck TT	–

Relation to earlier work. Inhomogeneous, spherically symmetric dust models with a centre are LTB solutions. Proposals to replace dark energy with a large local void, and the tests they faced, are directly relevant:

- An off-centre observer in a Gpc-scale void must sit within a few tens of Mpc of its centre to keep the CMB dipole small (Alnes & Amarzguioui 2006; Foreman, Moss, Zibin & Scott 2010).
- Large voids are constrained by the kinetic Sunyaev–Zel'dovich effect and by combined CMB, BAO and SN data (García-Bellido & Haugbølle 2008; Biswas, Notari & Valkenburg 2010; Moss, Zibin & Scott 2011; Zhang & Stebbins 2011).
- Copernican-principle tests apply generically (Clarkson, Bassett & Lu 2008).

The CoE configuration differs from those voids in two ways. The observer is far from the centre, and the CMB dipole is cancelled not by placement near the centre but by compensating structure inside the last-scattering sphere (§5.4). Whether that evades the Alnes–Foreman bounds depends on how physical the compensating structure is, which we discuss in §6. The kSZ and distance tests are not yet applied.

Status of the dynamics. Earlier revisions used Newtonian free fall from the CoE, with superluminal bulk velocities and no consistent rule for light propagation. In this revision, the background is the LTB solution (family "famP") that earlier fits already used, and every number is taken from it: H_0, age, the last-scattering radius and the observer's shell history (§2.2). The moving-lens kernel of the CMB field is still a Newtonian-gauge expression evaluated on that background, and its velocity frame remains open (§5.9).

Method. The density law is solved for from the data rather than assumed, in the sense of Ellis & Stoeger (1987), and the residuals left by each step are examined for structure (Tukey 1977; Box 1976; Mayo 1996). Because that procedure uses the same data to suggest and to fit each new term, §3.4 and Appendix A count every tuned quantity.

Layout. Section 2 sets out the model and the background, §3 the methods, §4 the data, §5 the results, §6 the tensions and limits, and §7 the conclusions. Symbols are listed in Table 2.

2 The model

2.1 The question and set-up

We separate *physical* expansion, the true radial motion of matter away from a centre of expansion, from *inferred* expansion, what an off-centre observer reconstructs from redshifts and luminosity distances under homogeneous–isotropic assumptions. The question is whether the second can look accelerated when the first is not. The geometry of Fig. 1 is the candidate.

Table 2: Parameters and symbols (values at the rev7 background).

Symbol	Meaning
D	observer–CoE distance, 34.96 Gpc (one value everywhere in rev7)
d	sampling radius of the CatWISE lock, 14.3005 Gpc (assumed; §5.1)
R, $R_{\rm ls}$, χ	source-sphere radius; $R = R_{\rm ls} = \eta_{\rm rec} D = 0.18957$ Gpc; distance from observer
H_0, t_0, V_O	LTB local expansion rate 34.31 km s^{-1} Mpc^{-1}; age 0.3138 Gyr; $V_O = H_0 D \approx 4.0c$
L, g; η, m_c	density scale length, shape; shape parameters of the η-edge law
l_0	Galactic longitude of the CoE axis, 19.87°
ψ; l, b	angle from the anti-CoE axis; Galactic longitude, latitude
A, B	normalisation and slope of $L_X \propto A T^B$
f	fraction of the field's $\ell = 2$ left uncancelled by the bodies, 0.26
lg A, n, α	source amplitude, tilt, running
A_w, r_s, ϕ, β	acoustic amplitude, scale, phase, amplitude tilt
k_D, m	damping cutoff and exponent
χ_w, s_w	radial window centre and width
B, k_B, r_2, ϕ_2	damped second mode
K_M, c_ℓ	moving-lens kernel; exact/Limber ratio
$D_\ell^{\rm fix}$	deterministic low-ℓ background (compensated)

Terms. **CoE**: centre of expansion, from which matter moves radially outward. **Shell and core**: the observer sits inside a thick shell of ordinary matter around the CoE; the core is the matter nearer the CoE. $\rho(r)$: matter density at distance r from the CoE, written $\rho_0 g(r/L)$. V_E: the expansion velocity of matter relative to the CoE; at the observer $\vec{V}_O$, at a galaxy $\vec{V}_g$. $V_s(\psi)$: the recession velocity measured along a line of sight, the local Hubble velocity $\vec{V}_H = \vec{V}_g - \vec{V}_O$ projected on it. ψ: angle between a line of sight and the observer–CoE axis; $\psi = 0$ points away from the CoE. Symmetry is axial about the CoE axis, so the sky anisotropy is an off-centre sampling of radial structure, not a direction-dependent scale factor.

2.2 One background: the LTB famP solution

Rev6 used three backgrounds: flat matter + radiation with $t_0 = 13.8$ Gyr (implying $H_0 = 47$), $\Omega_m = 0.3$, $H_0 = 70$ in the CatWISE lock, and ΛCDM with cold dark matter for the cosmic web. They cannot all hold. Rev7 uses one: the LTB famP solution, with the observer at distance D from the centre.

Choice of D. The Migkas profile was refitted at $D = 30$, 34.96, 42, 50 and 58.2 Gpc, with η, m_c and l_0 free and L locked (Fig. 2). The χ^2 values (independent-error, descriptive) are 9.528, 9.250, 9.594, 10.277 and 10.886. We adopt $D = 34.96$ Gpc. At that D, $\eta = 0.030$ sits at the edge of its allowed

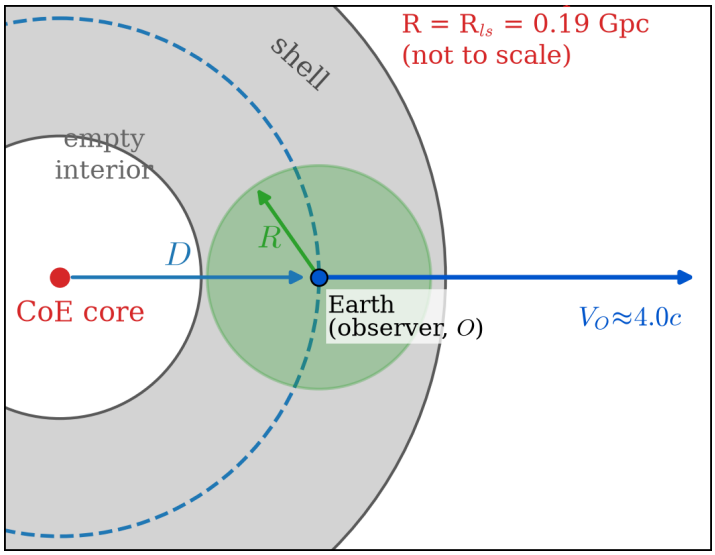


Figure 1: [**regenerated for rev7**] Set-up: the observer O inside a thick ordinary-matter shell at distance $D = 34.96\,\mathrm{Gpc}$ from the CoE, with the last-scattering sphere $R = R_{\mathrm{ls}} = 0.19\,\mathrm{Gpc}$ (not to scale) and $V_O = H_0 D \approx 4.0c$ from the LTB solution. The rev6 label "core $r_s = 163\,\mathrm{Gpc}$" had no source and is removed. The rev6 top panel (Hubble-vector sketch from the preprint, no script) is not carried over. Script: `Fig01b_geometry.py`.

range, so the minimum is bounded by the prior, and the profile is shallow (the 58.2 Gpc working value of rev6 is worse by $\Delta\chi^2 = 1.6$, which is not significant once the bins are correlated, §5.2).

Background relations. In the famP solution, lengths scale with D and times with D/c. In units of D/c, the observer's shell has age $t_0 = 0.002752$, local Ht_0 fixed, conformal time of recombination $\eta_{\mathrm{rec}} = 0.005422\,D$, and local expansion rate $H = 4.0006\,c/D$. At $D = 34.96\,\mathrm{Gpc}$ these give

$$H_0 = 34.31\ \mathrm{km\,s^{-1}Mpc^{-1}}, \qquad t_0 = 0.3138\ \mathrm{Gyr}, \qquad R_{\mathrm{ls}} = \eta_{\mathrm{rec}} D = 0.18957\ \mathrm{Gpc}, \tag{1}$$

with $V_O = HD \approx 4.0c$ and local density parameter $\Omega \approx 2.0 \times 10^4$ today (Fig. 3

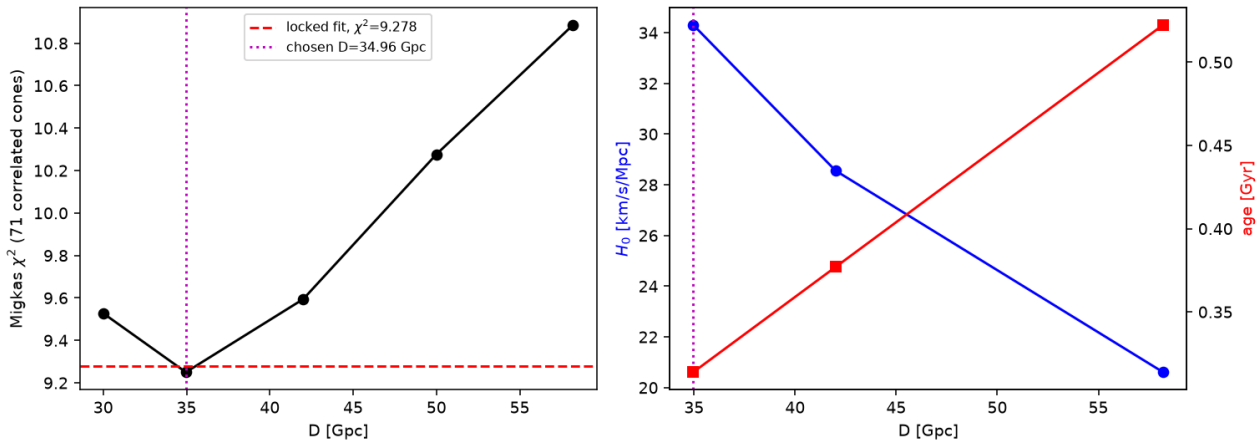


Figure 2: [**NEW in rev7**] Choice of background. Left: Migkas χ^2 (profile refit, CatWISE lock) against D; minimum at $D = 34.96\,\mathrm{Gpc}$, with η at the edge of its range. Right: LTB-implied H_0, age and R_{ls} against D. Differences in χ^2 are descriptive (correlated bins). Script: `N1_D_profile.py`.

). These follow from the solution; none was tuned.

Tension, stated. These values are not those of our universe ($H_0 \approx 70$, $t_0 \approx 13.8\,\mathrm{Gyr}$). They are internally consistent within the toy model, which is the point of this revision, but they show that the famP family with the observer on this shell is not yet a realistic cosmology. A further tension sits inside the cluster fit: the Migkas profile core corresponds to an age of 2.78 Gyr for the same D, against 0.314 Gyr for the LTB shell. Matching $t_0 = 13.8\,\mathrm{Gyr}$ at fixed Ht would need $H_0 \approx 0.78\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ on the LTB side. Removing these tensions requires a different LTB family (a different bang-time or curvature profile), which is listed as work in §6.

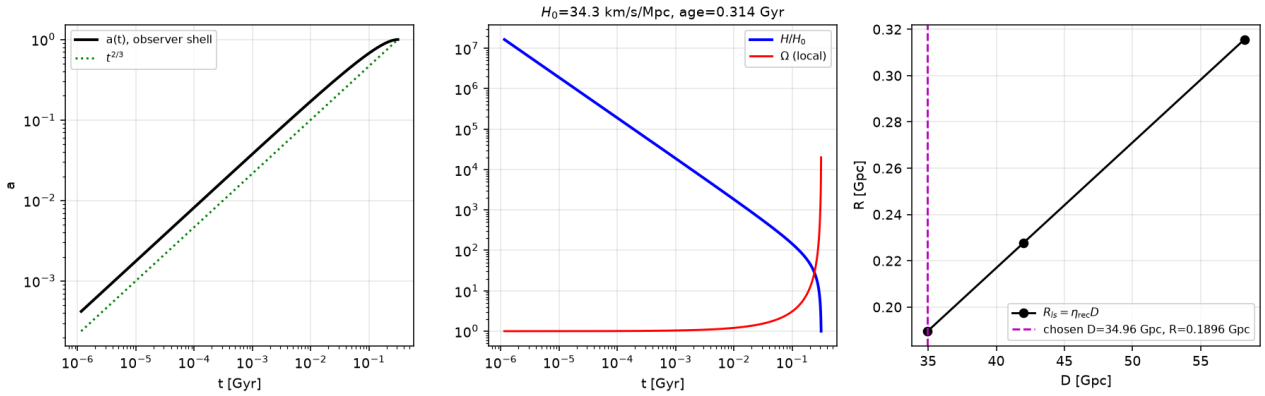


Figure 3: [**regenerated for rev7**] Background used everywhere in rev7. Left: scale factor on the observer's shell (LTB famP) against $t^{2/3}$. Middle: H/H_0 and the local Ω; $H_0 = 34.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, age 0.314 Gyr. Right: $R_{\mathrm{ls}} = \eta_{\mathrm{rec}} D$ against D; the chosen $D = 34.96\,\mathrm{Gpc}$ gives $R = 0.1896\,\mathrm{Gpc}$. Replaces the rev6 radiation + matter figure, which assumed $t_0 = 13.8\,\mathrm{Gyr}$. Script: `d4_background.py`.

Volumes and growth. A sphere of radius r has $V = \frac{4}{3}\pi r^3$. Today the observer's last-scattering sphere has $V_R = 0.0285\,\mathrm{Gpc}^3$, and the ratio $V_R/V_{D+R} = 1.57 \times 10^{-7}$ ($1.57 \times 10^{-5}\,\%$). Because the famP solution grows homologously on the observer's shell, all lengths scale with the same $a(t)$, and that ratio is constant in time (Fig. 4, middle). The volume growth rate is $\dot{V}/V = 3H$, close to $2/t$ (matter era) until the late-time departure in the bottom panel.

Validity. The radiation era and the recombination physics are still taken as FRW-like on the shell. A full LTB thermal history is not computed.

3 Methods

3.1 The inverse approach

The off-centre geometry is fixed. Equation (5) is the forward map from the density law to the cluster sky, and the line-of-sight kernels of §5.5 map the matter around the observer to the CMB. The data do not fix either uniquely, so each analytic density law is a trial form and the CatWISE lock removes one degree of freedom from each. After each fit the residual is inspected, and where it shows structure a further term is proposed. In this revision the inverse problem is used only in this light sense; no regularised inversion is attempted.

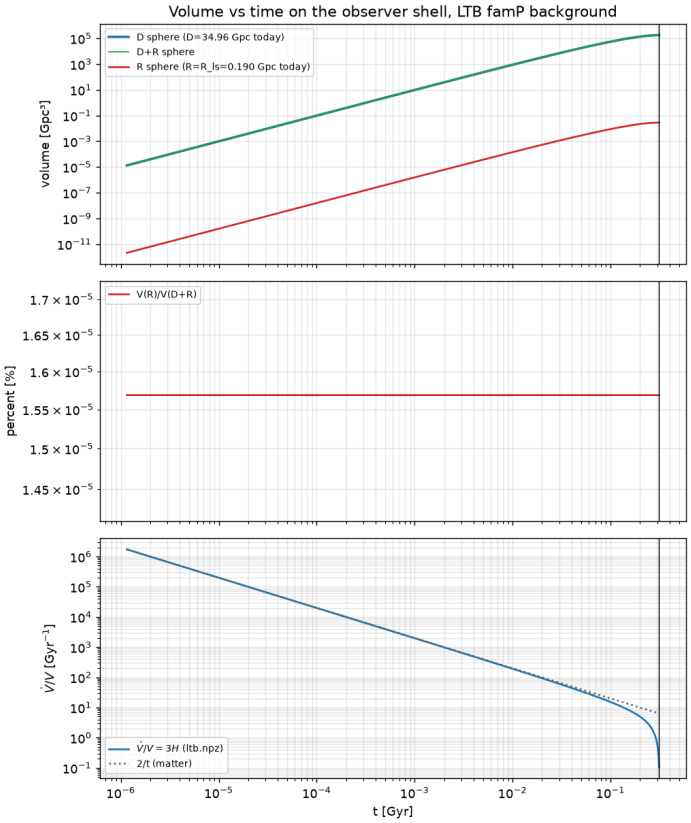


Figure 4: **[regenerated for rev7]** Volume against time on the observer's shell in the LTB famP background. Top: spheres of radius D, $D+R$ and $R = R_{\rm ls}$. Middle: $V(R)/V(D+R) = 1.57 \times 10^{-5}\,\%$, constant because the growth is homologous (the rev6 middle panel had an incorrect time dependence). Bottom: $\dot{V}/V = 3H$ against $2/t$. Script: `f02_volume_time.py`.

3.2 Spherical harmonics and D_ℓ

A sky map is expanded as $\Theta(\hat{n}) = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\hat{n})$. A mode of degree ℓ has angular half-wavelength $\theta \approx 180^\circ/\ell$. For a statistically isotropic sky, $\langle a_{\ell m} a^*_{\ell' m'} \rangle = C_\ell \delta_{\ell\ell'} \delta_{mm'}$, and

$$D_\ell \equiv \frac{\ell(\ell+1)}{2\pi} C_\ell T_0^2 \quad [\mu\mathrm{K}^2], \qquad T_0 = 2.7255 \times 10^6\,\mu\mathrm{K}. \tag{2}$$

If the $a_{\ell m}$ are Gaussian, $(2\ell+1)\hat{C}_\ell/C_\ell$ is $\chi^2_{2\ell+1}$-distributed, so $\Delta C_\ell/C_\ell = \sqrt{2/(2\ell+1)}$ (63 % at $\ell = 2$, 6.7 % at $\ell = 220$). This is how one-sky realisations are drawn in §5.7.

3.3 Goodness of fit

For the CMB, with Planck's asymmetric low-ℓ errors,

$$\chi^2 = \sum_\ell \frac{(m_\ell - d_\ell)^2}{\sigma_\ell^2}, \qquad \sigma_\ell = \begin{cases} \sigma_\ell^+ & m_\ell > d_\ell \\ \sigma_\ell^- & m_\ell \le d_\ell. \end{cases} \tag{3}$$

Two CMB statistics are kept apart. The *fit objective* is unbinned $\ell = 2$–29 plus Planck's binned spectrum above $\ell = 30$; it was minimised. The *reported* statistic is unbinned $\ell = 2$–2500; it was not. Both appear in Table 5. Pearson r^2 is given only as a shape descriptor and is not used for comparison.

For the Migkas profile, the independent-error χ^2 uses the mean of the asymmetric errors. Because the 71 cones overlap, it is not a valid goodness-of-fit statistic. Section 5.2 gives an approximate covariance-aware version.

3.4 Model comparison and parameter accounting

Models are compared by $\Delta\chi^2$ against a reference with the same data and likelihood, and by

$$\mathrm{AIC} = \chi^2 + 2p, \qquad \mathrm{BIC} = \chi^2 + p \ln N, \tag{4}$$

where p counts every quantity chosen with the data in view, including discrete choices. Appendix A lists p for each claim. Where a claim rests on a pattern (the anomaly score of earlier revisions), its significance must come from a null distribution generated by the same pipeline. In rev7 that score is withdrawn (§5.10).

4 Data

Cluster scaling relations. Migkas et al. (2020) test isotropy with the L_X–T relation of 313 clusters. L_X depends on luminosity distance and T does not, so a directional change in the normalisation A traces a directional distance or expansion scale: with AH_0^2 fixed, $H/H_{\rm ref} = 1/\sqrt{A/A_{\rm all}}$. $A/A_{\rm all}$ is given in sliding cones of width $\Delta l = 90°$ against Galactic longitude (their Fig. 6), digitised here at 71 points every 5°. Adjacent points share most of their clusters. [TO ADD: digitisation method and its estimated error.] The 313-cluster catalogue itself is not on the working machine, so a per-cluster likelihood is not possible in this revision.

CatWISE dipole (Secrest et al. 2021): $\mathcal{D} = 0.01554$, of which ≈ 0.007 is the expected kinematic part. It is used only through the lock of §5.1.

Planck 2018 *TT* (Planck Collaboration 2020a): D_ℓ for $\ell = 2$–2500, asymmetric errors at low ℓ, binned file for $\ell \geq 30$. The ΛCDM curve is the Planck best fit.

5 Results

5.1 Density laws, the CatWISE lock and the Migkas fit

Line of sight. Put the CoE at the origin and the observer at $\vec{D}$. A point at distance s in direction $\hat{n}$ has $r(s,\hat{n}) = \sqrt{D^2 + s^2 + 2Ds\cos\psi}$. Averaging a response f over the cluster depth distribution $p(s)$,

$$\frac{A}{A_{\rm all}}(\hat{n}) = \frac{\int ds\, p(s)\, f[\rho(r(s,\hat{n}))]}{\langle \int ds\, p(s)\, f[\rho]\rangle_{\rm sky}}. \tag{5}$$

To first order in s/L this is a dipole in $\cos\psi$ with amplitude set by $(\rho'/\rho)_D$. [unverified: the exact f and $p(s)$ in the Migkas scripts were not re-derived.]

CatWISE lock. With $\rho(r) = \rho_0 g(r/L)$ and $\rho(D) = 0.3\rho_{\rm crit}$, require

$$\delta \equiv \frac{\rho(D-d) - \rho(D+d)}{\rho(D)} = 0.031 \approx 2\mathcal{D}. \tag{6}$$

For $g = e^{-x}$, $\delta = 2\sinh(d/L)$, so

$$L = \frac{d}{\operatorname{arcsinh}(\delta/2)} = \frac{14.3005}{0.0154994} = 922.65 \text{ Gpc}, \tag{7}$$

independent of D. With the exact $2\mathcal{D} = 0.03108$, $L = 920.3$ Gpc; the code uses 0.031 throughout. For the other laws Eq. (6) is solved numerically at $D = 34.96$ Gpc, with the shape parameters held fixed (Table 3, Fig. 5).

Table 3: CatWISE-locked laws at $D = 34.96$ Gpc (rev6: 58.2 Gpc). Source: `d1_migkas_laws.json`.

law	$g(x)$	L [Gpc]	ρ_0/ρ_c
exp	e^{-x}	922.65	0.3116
power	$(1+x)^{-1.2}$	1072.4	0.3118
cored	$(1+x^2)^{-1}$	250.8	0.3058
Gauss	$e^{-x^2/2}$	179.3	0.3058
soft	$(1+x^{0.3})^{-1}$	25715	0.3414
stretched	$e^{-x^{0.25}}$	78314	0.3469

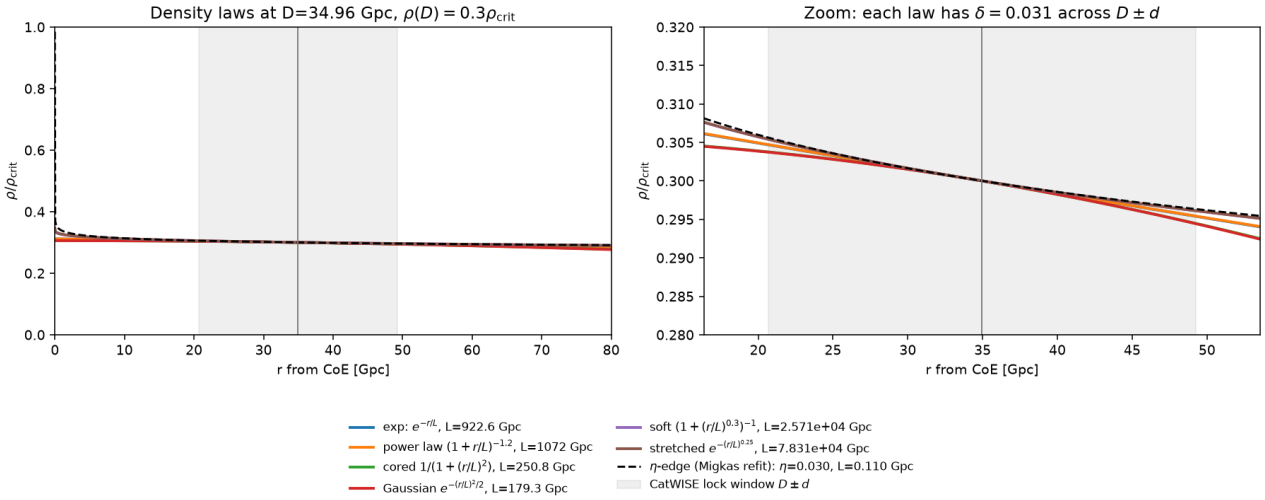


Figure 5: [**regenerated for rev7**] CatWISE-locked density laws at $D = 34.96$ Gpc: $\rho/\rho_{\rm crit}$ against distance from the CoE (left, to 80 Gpc) and zoom on $D \pm d$ (right), with the η-edge law of the Migkas refit. Inside the window the laws are indistinguishable; they differ only near the CoE, where no data constrain them. Script: `d1_migkas_laws.py`.

Open assumptions of the lock. (1) The full observed dipole is attributed to density; the kinematic part (≈ 0.007, and in this model also V_O) has not been subtracted. (2) Source bias, magnification bias and the CatWISE redshift kernel do not enter. (3) $d = 14.3$ Gpc is not the depth of either sample, and L depends strongly on it ($L = 193.6$ Gpc at $d = 3$ Gpc for the exponential). On the rev7 background, d is also much larger than $R_{\rm ls} = 0.19$ Gpc, so the lock samples radii that no photon reaching us has crossed. [TO REDO: lock with the kinematic-subtracted dipole and the CatWISE dN/dz mapped onto the LTB distance–redshift relation.]

Fit (priority 1, locked). The free-η profile at $D = 34.96$ Gpc gives $\chi^2 = 9.250$ ($r^2 = 0.749$) with $\eta = 0.030$, $m_c = 4.337$, $l_0 = 19.87°$, $L = 0.110$ Gpc. The curve carried through the rest of the

paper is the locked η-edge fit, $\chi^2 = 9.278$ ($r^2 = 0.748$). That fit is held fixed. Every later step (the compensating bodies, the CMB field) is checked against it (Fig. 6).

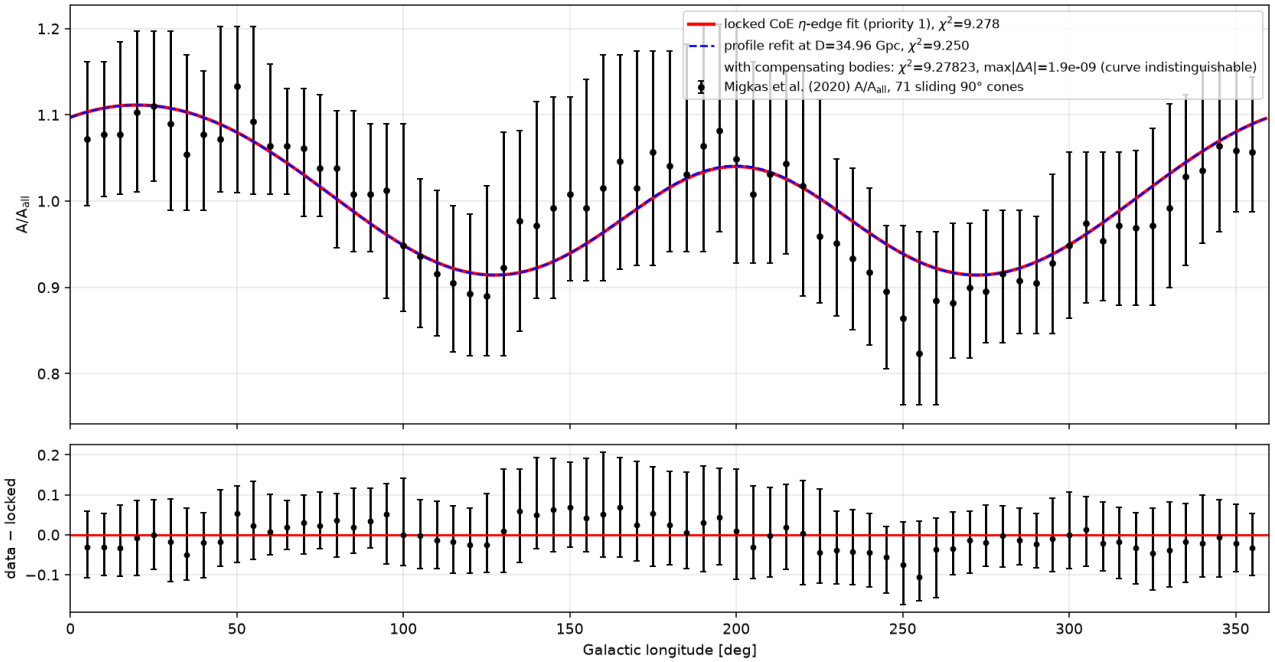


Figure 6: [**regenerated for rev7**] Migkas $A/A_{\rm all}$ against Galactic longitude (71 sliding 90° cones, strongly correlated). Red: locked CoE η-edge fit, $\chi^2 = 9.278$. Blue dashed: profile refit at $D = 34.96\,$Gpc, $\chi^2 = 9.250$. The curve with the compensating bodies of §5.4 added ($\chi^2 = 9.27823$, $\max|\Delta A| = 1.9 \times 10^{-9}$) is indistinguishable. Bottom: data minus locked fit. χ^2 values are independent-error and descriptive. Script: `Fig04_amp_full.py`.

5.2 Correlated bins and model comparison for the cluster profile

Adjacent cones share most of their clusters. If clusters are roughly uniform in longitude, two 90° cones separated by Δl share a fraction

$$\rho(\Delta l) = \max\left(0,\, 1 - \frac{|\Delta l|}{90^\circ}\right) \tag{8}$$

of their clusters (circular in l). We take this as the correlation matrix and scale it by the errors. The 71×71 matrix is nearly singular, so we use its pseudo-inverse keeping eigenvalues above 1 % of the largest (15 modes). The effective number of independent bins, $n_{\rm eff} = (\mathrm{tr}\, C)^2/\mathrm{tr}\, C^2$, is 5.98.

Table 4: Migkas $A/A_{\rm all}$: independent-error and overlap-covariance χ^2, k fitted parameters. p-values are for $\Delta\chi^2_C$ against isotropy with k degrees of freedom. Source: `N2_migkas_cov.json`.

model	k	$\chi^2_{\rm indep}$	χ^2_C	$\Delta\chi^2_C$ vs iso	p	BIC_C
isotropy	0	45.14	8.99	–	–	–
free dipole	3	29.36	6.90	2.09	0.55	12.26
dipole + quadrupole	5	10.86	5.20	3.79	0.58	14.14
CoE, $D = 34.96$ (D, η, m_c, l_0)	4	9.25	3.88	5.11	0.28	11.04

Reading. The ordering is the same with and without the covariance: the CoE geometry gives the lowest χ^2 with fewer parameters than a dipole + quadrupole, and the lowest BIC. But the improvements over isotropy are not significant ($p = 0.28$–0.58) once the bins are correlated, because the 71 cones carry about six independent numbers. The cluster result is therefore qualitative: the off-centre geometry produces the two-hump shape in the right places, with an amplitude tied to CatWISE, but these data alone cannot reject isotropy. The covariance model (Eq. 8) is an approximation. [TO COMPUTE: bootstrap covariance from the 313-cluster catalogue.]

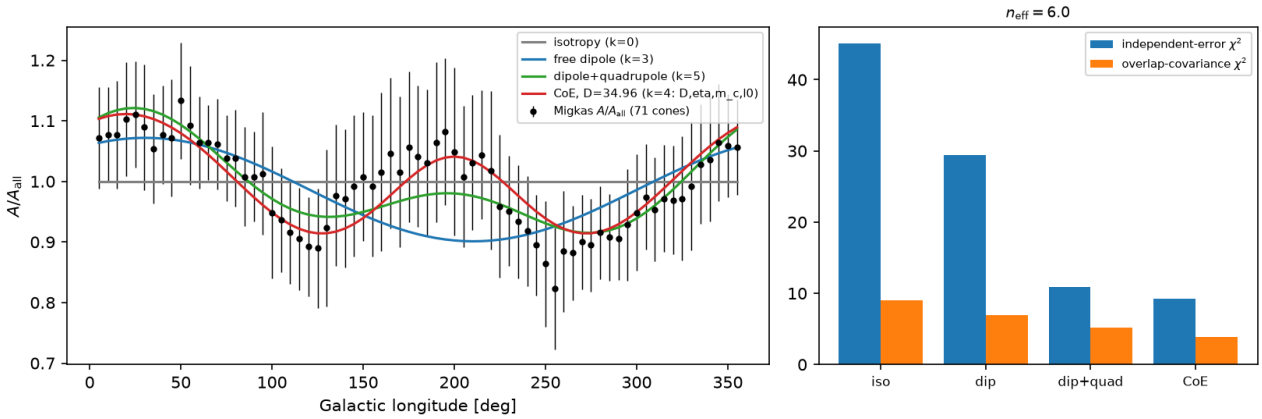


Figure 7: [**NEW in rev7**] Left: Migkas $A/A_{\rm all}$ with isotropy, a free dipole ($k = 3$), a dipole + quadrupole ($k = 5$) and the CoE profile at $D = 34.96\,$Gpc ($k = 4$). Right: independent-error and overlap-covariance χ^2 for each ($n_{\rm eff} = 6.0$). Script: `N2_migkas_cov.py`.

$B/B_{\rm all}$. Rev6 reproduced B only with a fitted longitude dipole whose spreadsheet inputs are not on the machine. B is not refitted on the rev7 background and is not claimed.

5.3 The CMB: the deterministic sky at last scattering

Carried to last scattering at $R = R_{\rm ls} = 0.18957\,$Gpc on the rev7 background, the smooth CoE sky (the background map of the famP solution at $D = 34.96\,$Gpc) has rms $\Delta T/T = 2.32 \times 10^{-5}$. Its multipole amplitudes are 2.18×10^{-5} at $\ell = 1$, 9.3×10^{-8} at $\ell = 2$, 7.7×10^{-6} at $\ell = 3$, and below 1.4×10^{-7} for $\ell = 4$–5. The map depends only weakly on D (the $D = 42$ map reproduces the rev5 one with correlation 0.99999999). This replaces the rev6 statement that a percent-level geometric multipole survives: at the consistent, much smaller $R_{\rm ls}$, the deterministic sky is already at the 10^{-5} level, dominated by the dipole and octupole.

5.4 Compensating structure inside the last-scattering sphere

The question of §1 is whether some physically allowed set of masses can make that sky as flat as observed. We place 3,000 point bodies (seed 2000) at $0.075R < r < 0.875R$ from the observer, softened at $0.025R$, each with $|M| \le 10^{14}\,M_\odot$, and solve a bounded least-squares problem for their masses. The target is to cancel the background at $\ell = 1$–12, while penalising $\ell = 13$–40, and to cancel a fraction $1 - f = 0.74$ of the stochastic field's own $\ell = 2$ on the seed-42 sky, so that the observed D_2 is approached.

Result (`p5_c_p8`). 975 clusters ($M_+ = 2.98 \times 10^{15} M_\odot$) and 904 voids ($M_- = -3.25 \times 10^{15} M_\odot$), counting bodies above threshold. No body sits at the mass bound. The residual sky has rms 5.67×10^{-6}

and maximum 1.27×10^{-5}. Its dipole is 3.9×10^{-11} and its octupole 1.1×10^{-9}. The relevant error is not the residual itself but the residual against the target. With $F_{\rm target}$ the $\ell = 2$ part of the field map scaled by $(1 - f)$, the maximum of $|\mathrm{Resid} + F_{\rm target}|$ over the sky is 1.50×10^{-6} (rms 3.2×10^{-7}), below the 10^{-5} goal. The $\ell = 2$ part of that error is 8.3×10^{-10} (`p5_err.py`). On the seed-42 sky, D_2 falls from 1,042.3 to 70.49 μK^2 (target 70.46).

Cluster sky unchanged. The bodies perturb the cluster lines of sight by $\Delta\Phi/c^2 = 2.8$–4.3×10^{-9}. The Migkas χ^2 moves from 9.278230532 to 9.278230704 ($r^2 = 0.74805$ unchanged, $\max|\Delta A| = 1.9 \times 10^{-9}$).

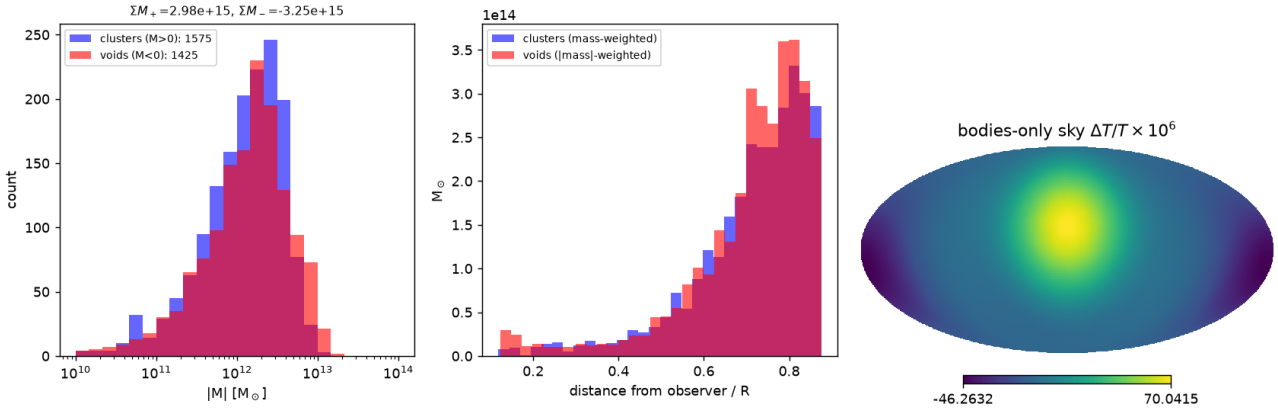


Figure 8: [**regenerated for rev7**] The compensating bodies of the final run (`p5_c_p8`). Left: mass histogram of clusters and voids. Middle: mass against distance from the observer in units of R. Right: the bodies-only sky $\Delta T/T$. Replaces the rev6 Press–Schechter/void-function web, which used ΛCDM σ_8, Ω_m and a CDM transfer function. Script: `d2_web.py`.

Validity. The bodies are Newtonian point masses with tidal-shear sky kernels inside a sphere of radius 0.19 Gpc. Their total masses are $\sim 3 \times 10^{15} M_\odot$ of each sign, comparable to a few rich clusters. Whether such a configuration forms naturally is not addressed, and a configuration solved for to cancel the dipole is, by construction, fine-tuned. It shows existence, not likelihood.

5.5 Stochastic field and sky projection

The density contrast is a zero-mean Gaussian random field with power $P_\delta(k)$, and $\Phi_k = -(4\pi G\bar\rho/k^2)\delta_k$. A photon crossing it picks up the Sachs–Wolfe, integrated Sachs–Wolfe and moving-lens terms. The moving lens supplies almost all of the power. Confined to a radial window $W(\chi) = \exp[-(\chi - \chi_w)^2/2s_w^2]$, the exact projection is

$$C_\ell = \frac{2}{\pi}\int k^2 dk\, P_\Phi(k)|\Delta_\ell(k)|^2, \qquad \Delta_\ell(k) = \int d\chi\, KW j_\ell(k\chi), \tag{9}$$

used for $\ell \le 50$, and the Limber form

$$D_\ell \simeq \frac{\ell(\ell+1)T_0^2}{2\pi}\int d\chi\, \frac{[K_M W]^2}{\chi^2}\,\frac{k^2}{3}P_\Phi(k)\Big|_{k=\nu/\chi}, \quad \nu = \ell + \tfrac{1}{2}, \tag{10}$$

is corrected by $c_\ell = C_\ell^{\rm exact}/C_\ell^{\rm Limber}$ above $\ell = 50$.

Changes on the rev7 background. The kernels previously had $D = 42\,\mathrm{Gpc}$ hard-coded in the moving-lens velocity factor and $R = 4\,\mathrm{Gpc}$. Both now use $D = 34.96$ and $R = R_{\mathrm{ls}} = 0.18957\,\mathrm{Gpc}$, with all lengths in $P(k)$ and W scaled by $s = R/4$ and wavenumbers by $1/s$ as the starting point. D_ℓ^{fix} is the low-ℓ ($\ell \le 40$) spectrum of the compensated sky of §5.4.

Source spectrum. The fitted form, with $K_0 = 100/s$ and k in Gpc^{-1}, is

$$P_\delta = 10^{\lg A} k^{\,n+\alpha \ln(k/K_0)} \Big[1 + A_w \Big(\tfrac{k}{K_0}\Big)^{\beta} \cos(k r_s + \phi) + B e^{-k/k_B} \cos(k r_2 + \phi_2)\Big] e^{-(k/k_D)^m}. \qquad (11)$$

The oscillatory terms are phenomenological and fitted. They describe the acoustic peaks; they do not derive them. With $\alpha = \beta = 0$ and $m = 2$ the form reduces to the rev6 12-parameter model. The 15 parameters were fitted by a multistart Nelder–Mead search (12 starts) of the fit objective, with the exact/Limber correction refreshed between passes (correlation of successive c_ℓ 0.899 then 0.978). Best fit: $\lg A = -10.508$, $n = 0.298$, $\alpha = -0.00365$, $A_w = 0.297$, $\beta = -0.00666$, $r_s = 0.003732\,\mathrm{Gpc}$, $\phi = 1.414$, $k_D = 4104\,\mathrm{Gpc}^{-1}$, $m = 1.250$, $\chi_w = 0.1724\,\mathrm{Gpc}$, $s_w = 0.01096\,\mathrm{Gpc}$, $B = 0.734$, $k_B = 3343\,\mathrm{Gpc}^{-1}$, $r_2 = 0.00188\,\mathrm{Gpc}$, $\phi_2 = 4.054$. Note $r_2 \approx r_s/2$, as in rev6, and $\chi_w = 0.91R$.

Peak spacing. An oscillation $\cos(k r_s)$ gives peaks spaced by $\Delta\ell \approx 2\pi\chi_{\mathrm{eff}}/r_s$. With $r_s = 0.003732\,\mathrm{Gpc}$ and $\chi_{\mathrm{eff}} \approx \chi_w = 0.1724\,\mathrm{Gpc}$, $\Delta\ell \approx 290$, close to the spacing of the model peaks in Table 5.

No parameter uncertainties. Nelder–Mead returns a point. [TO COMPUTE: Hessian or MCMC widths at the optimum.]

5.6 TT fit

Table 5: TT statistics of the current solve (p8 field on the rev7 background, with the compensating bodies). Δ values are against ΛCDM with $N = 2499$, $p = 15$ vs 6. Sources: `p6_p8.json`, `N4_tt_stats.json`.

	χ^2_{2-30}	$\chi^2_{30-2500}$	χ^2_{2-2500}	r^2_{2-2500}	$\Delta\chi^2$	ΔBIC
ΛCDM (Planck best fit)	20.14	2538.39	2558.49	0.983	–	–
CoE ensemble mean	29.33	2586.91	2616.19	0.982	+57.7	+128.1
CoE seed 42	32.76	3308.41	3340.22	–	+781.7	+852.1
CoE seed 42, compensated $\ell \le 40$	31.38	3308.17	3338.67	0.975	+780.2	+850.6

Fit objective (low-ℓ unbinned + binned, the minimised quantity): CoE 141.78, ΛCDM 98.75. Peaks, with one finder for all three spectra (Savitzky–Golay 61/3, prominence 20, separation ≥ 200): Planck 224, 514, 822, 1124, 1405; CoE 221, 536, 814, 1126, 1417; ΛCDM 220, 536, 813, 1127, 1421.

Reading. On the consistent background the ensemble mean comes within $\Delta\chi^2 = +58$ of ΛCDM over $\ell = 2$–2500 (rev6, inconsistent background: +1,229). Because p is larger, ΔBIC is +128, so ΛCDM is still strongly preferred. The model peak positions agree with ΛCDM's to within 1–4 in ℓ; on unbinned data the smoothed Planck second peak (514) is pulled by noise, so the peak list is descriptive. The seed-42 numbers are one draw. Section 5.7 puts them in context.

5.7 Distribution of skies

The path-dependence argument of §1 makes the distribution of one-sky χ^2 the relevant statistic. We draw 1,000 skies as $D_\ell = D_\ell^{\mathrm{fix}} + D_\ell^{\mathrm{field}} \chi^2_{2\ell+1}/(2\ell+1)$ and, for a like-for-like comparison, 1,000 ΛCDM

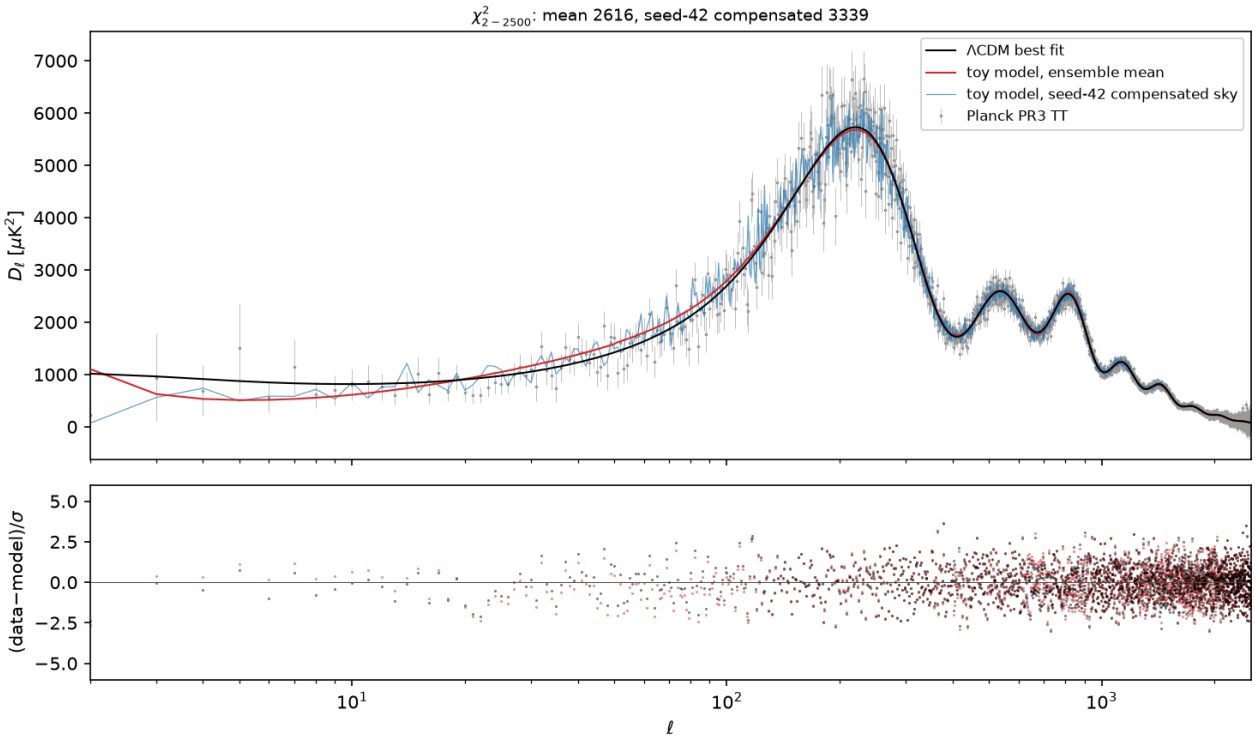


Figure 9: [**regenerated for rev7**] TT spectrum of the current solve against Planck 2018 and ΛCDM, with residual panel. Models are drawn above the data. Ensemble mean $\chi^2_{2-2500} = 2{,}616$; compensated seed-42 sky 3,339; ΛCDM 2,558. Script: `d3_stoch.py`.

skies the same way. Planck's errors already include cosmic variance, so absolute one-sky χ^2 values double-count it for both models. Only the comparison between the two distributions is meaningful.

Results (`N5_seed_distribution.json`). Over $\ell = 2$–2500 the CoE median is 3,347 (16–84 %: 3,286–3,408), against 3,279 (3,221–3,338) for ΛCDM. Over $\ell = 2$–30 the CoE median is 50.1 (39.3–63.8), against 42.4 (32.0–55.8). Seed 42 (3,340; 32.8) is typical of the model distribution. Of the CoE skies, 26 % pass a strict 5 % consistency cut at low ℓ ($\chi^2_{2-30} < 43$), and 99.3 % pass the loose rev5 rule (< 100). This answers the pass-rate question (referee M9) for the current field. The rev6 Monte Carlo web ensemble is not re-run.

5.8 Joint summary

Figure 11 collects the pieces of the solve on the one background.

5.9 Open CMB problems

1. Which velocity. The moving-lens kernel uses the CoE radial velocity of the matter. In a Newtonian picture, a velocity common to observer, lens and source produces no anisotropy; only the velocity relative to the observer, $\vec{V}_H = \vec{V}_g - \vec{V}_O$, should enter. Near the observer $\vec{V}_H \approx H\chi\hat{n}$ is nearly radial, and a radial velocity contributes little to the transverse (Limber) signal. On the rev7 background $R_{\rm ls} = 0.19$ Gpc and $V_H \lesssim HR \approx 0.02c$, against $V_O \approx 4c$, so the relative-velocity kernel would be about two orders of magnitude weaker in amplitude. The amplitude $\lg A$ would absorb a uniform rescaling, but the shape would change. [TO COMPUTE: refit with the relative-velocity kernel.]

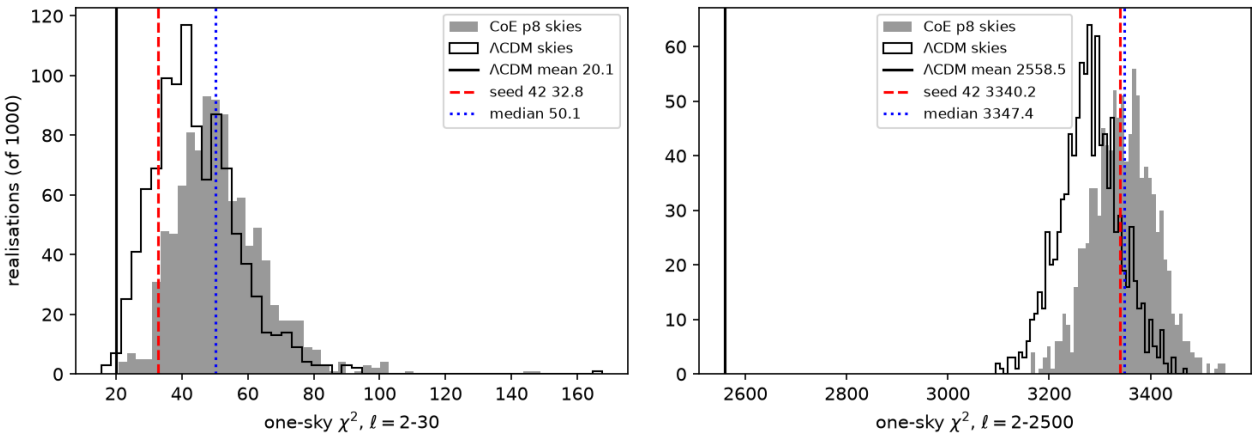


Figure 10: [**NEW in rev7**] One-sky χ^2 over 1,000 realisations of the current CoE field (grey) and of ΛCDM drawn the same way (black outline), for $\ell = 2$–30 (left) and 2–2500 (right). Vertical lines: ΛCDM mean curve, seed 42, CoE median. Script: `N5_seed_distribution.py`.

2. Statistical anisotropy. With $\vec{u}$ along a fixed axis, the local small-scale power varies as $1 - \cos^2\theta = \frac{2}{3}[1 - P_2(\cos\theta)]$: a 100 % quadrupolar modulation that vanishes along the CoE axis and is hidden by the orientation average $\langle(\hat{k}\cdot\hat{u})^2\rangle = 1/3$. Planck constrains such modulations at the percent level (Planck Collaboration 2020b). If the relative velocity is used instead, the modulation changes. [TO COMPUTE: predicted modulation and comparison with Planck.]

3. Polarisation. TE and EE are not computed. They are the natural held-out test (§6).

5.10 Sky anomalies

Rev6 reported that the A and B residuals, after harmonics $n = 1$–3, agreed in sign with 11 of 12 anomalies. The audit showed that the A reference curve was a Fourier fit to the data, the spots are paired, the orientation was chosen post hoc, and no script for the maps or scores survives. On the rev7 background the residual maps have not been rebuilt, so the score is *withdrawn* from this revision rather than repeated. [TO COMPUTE: rotation and isotropic-mock null with the rev7 profile, pairs collapsed.]

6 Discussion

What is fitted and what is predicted. Predicted before the data: the two-hump shape, ring locations, the sign of the gradient (Table 1). Fitted: D, η, m_c, l_0; L by the lock; the masses of the compensating bodies; the 15 source-spectrum parameters. Appendix A counts them.

Cluster sky. A single off-centre axis with a CatWISE-locked gradient reproduces the two-hump shape of $A/A_{\rm all}$ with fewer parameters than a dipole + quadrupole and with the lowest BIC, but with about six independent bins no model is significantly preferred over isotropy. The direction is right; the significance is qualitative.

CMB. On one background, a physically bounded set of bodies cancels the deterministic dipole and octupole below 10^{-9}, reaches the D_2 target within 1.5×10^{-6}, and leaves the cluster fit untouched. The stochastic field brings the ensemble-mean TT fit to within $\Delta\chi^2 = +58$ of ΛCDM. That is large

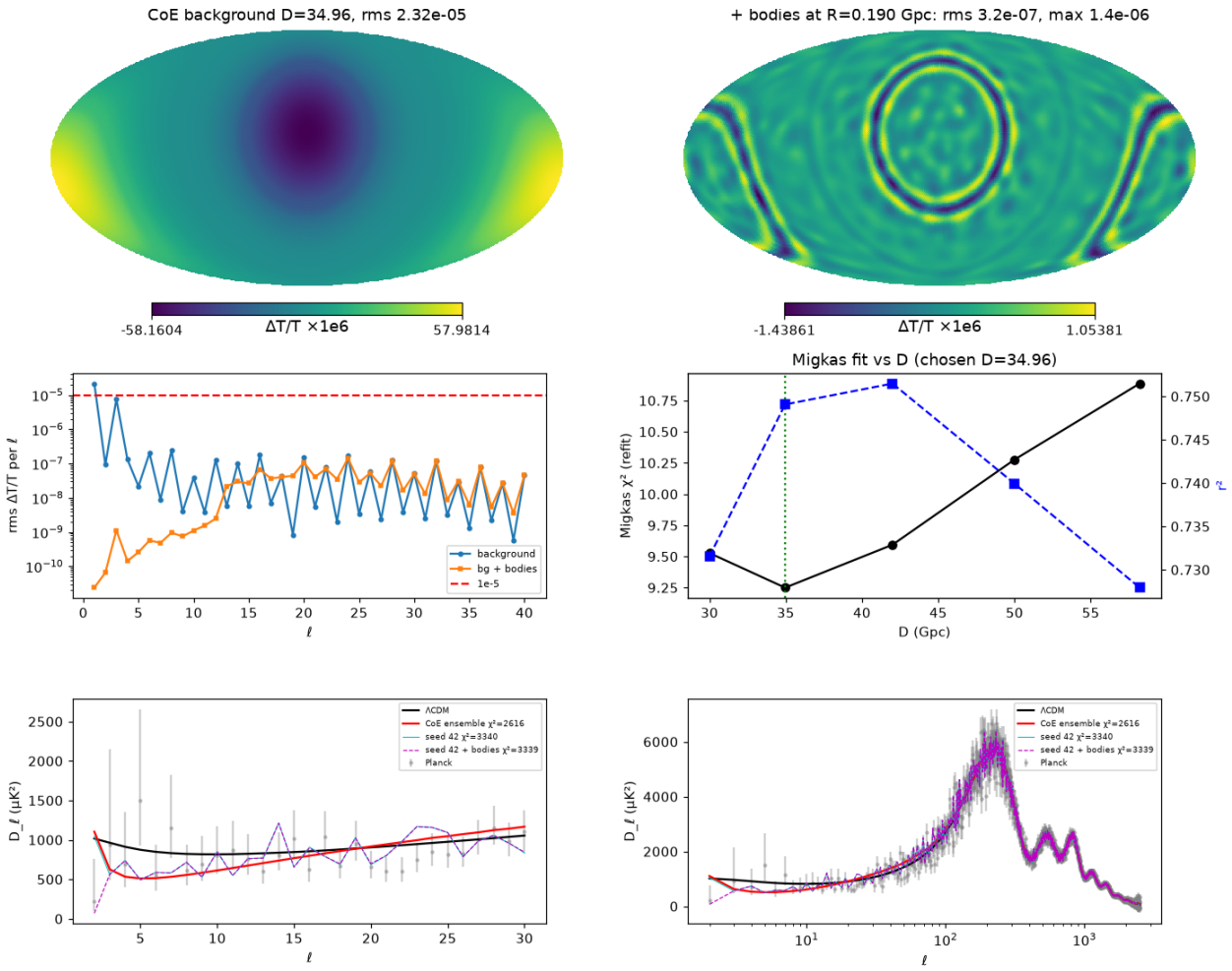


Figure 11: **[NEW in rev7]** Joint summary of the rev7 solve ($D = 34.96\,\mathrm{Gpc}$, $R = 0.18957\,\mathrm{Gpc}$). Top: deterministic background sky (rms 2.32×10^{-5}); residual after a background-only compensation run (bodies at R, $f = 1$; rms 3.2×10^{-7}). Middle: rms per ℓ before and after that run; Migkas χ^2 against D. Bottom: low-ℓ and full TT spectra of the final run (p8 field, `p5_c_p8` bodies) with ΛCDM and Planck. Script: `N3_joint_summary.py`.

progress from rev6 (+1,229), but $\Delta\mathrm{BIC} = +128$ and the like-for-like realisation medians differ by ≈ 70, so ΛCDM fits better. The acoustic peaks are put in through P_δ, and the moving-lens velocity frame may change the result qualitatively.

Off-centre bounds. Alnes & Amarzguioui (2006) and Foreman et al. (2010) bound an off-centre observer in a void by the CMB dipole and quadrupole *without* compensating structure. Here the dipole is removed by structure inside R_{ls}. This is a loophole only if that structure is plausible. A solved-for configuration is fine-tuned by construction, so the bound is not evaded in any probabilistic sense.

Internal consistency. Rev6 listed two observer distances, three backgrounds, three axis longitudes and two orientations. Rev7 uses one distance ($D = 34.96\,\mathrm{Gpc}$), one background (LTB famP), and one axis ($l_0 = 19.87°$). The sky-orientation sequences of rev6 are not carried over. Remaining tensions: $H_0 = 34.3$ and $t_0 = 0.314\,\mathrm{Gyr}$ against observed values, and the profile-core age (2.78 Gyr) against the LTB age.

Limits.

- The thermal history at recombination is FRW-like by assumption.
- The moving lens is a Newtonian-gauge kernel on an LTB background, with the velocity frame open.
- The cluster covariance is a geometric approximation, not a bootstrap.
- The CatWISE lock ignores the kinematic dipole, bias and the redshift kernel, and samples radii beyond $R_{\rm ls}$.
- No parameter uncertainties are given for the 15-parameter source spectrum.
- Whether an off-centre observer would infer apparent acceleration from supernova distances has not been computed.

Next steps, in priority order.

1. Refit the field with the relative-velocity moving-lens kernel; compute the quadrupolar modulation and compare with Planck.
2. Find an LTB family that keeps the cluster and CMB results with H_0 and t_0 nearer observed values.
3. Redo the CatWISE lock with the kinematic-subtracted dipole and the CatWISE dN/dz on the LTB distance–redshift relation.
4. Bootstrap the Migkas covariance from the 313-cluster catalogue.
5. Held-out tests: predict TE/EE; fit $\ell \leq 1200$ and predict $\ell > 1200$; fit A and predict B.
6. Rebuild the anomaly maps on the rev7 profile with a rotation null.
7. Compute $D_L(z, \hat{n})$ along off-centre lines of sight against Pantheon+.

7 Conclusions

1. The toy universe now runs on one background: an LTB solution with the observer at $D = 34.96\,\mathrm{Gpc}$, giving $H_0 = 34.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, age $0.314\,\mathrm{Gyr}$ and $R_{\rm ls} = 0.190\,\mathrm{Gpc}$. Those values are a tension of the model, stated rather than hidden.
2. On the cluster sky the off-centre geometry goes the right way. It gives the lowest χ^2 and BIC among isotropy, dipole, dipole + quadrupole and CoE under an approximate covariance. With about six effective bins, the preference is qualitative.
3. On the CMB, a bounded set of 975 clusters and 904 voids inside the last-scattering sphere flattens the deterministic CoE sky to the target within 1.5×10^{-6} without disturbing the cluster fit. With a phenomenological source spectrum, the TT ensemble mean comes within $\Delta\chi^2 = +58$ of ΛCDM, while $\Delta\mathrm{BIC} = +128$ still favours ΛCDM.
4. The decisive open problems are the moving-lens velocity frame and the quadrupolar modulation it implies, a realistic H_0 and age, and a held-out prediction.

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A Parameter accounting

Claim	Quantities chosen with the data in view	p (approx.)
Migkas $A/A_{\rm all}$ (§5.1)	D, η, m_c, l_0; L by the lock; law family (1 of 7)	4 continuous + 1 discrete
Choice of D (Fig. 2)	profile minimum over 5 values	included above
CatWISE lock	$d = 14.3005\,$Gpc, $\delta = 0.031$ (assumed)	0 fitted, 2 assumed
Background (§2.2)	LTB famP family; D from the cluster fit	1 discrete
Compensating bodies (§5.4)	3,000 masses (bounded LSQ), positions seeded (2000); $\ell_{\max} = 12$, $\lambda = 10^{-5}$, $f = 0.26$	solved, not fitted to TT; f 1
TT field (§5.5)	$\lg A, n, \alpha, A_w, \beta, r_s, \phi, k_D, m, \chi_w, s_w, B, k_B, r_2, \phi_2$	15
TT structural choices	acoustic term; damped term (period first read off a residual in rev5); running and tilt terms added in rev7	≥ 3 discrete
Anomaly score	withdrawn in rev7	–

B Figure provenance

All scripts are in `fig7/` beside their PNG and read the solve outputs in `joint_solve/phase2/`.

Fig.	file	status	inputs
1	Fig01b_geometry	regenerated (labels to rev7)	p3/ltb.npz
3	d4_background	regenerated (replaces rad+matter)	ltb.npz, tab.npz, p3 json
4	f02_volume_time	regenerated (middle panel fixed)	ltb.npz
5	d1_migkas_laws	regenerated at $D = 34.96$	lock solver in script; p1
6	Fig04_amp_full	regenerated	Migkas digitised; p1; p7
2	N1_D_profile	**NEW**	p1_migkas_D*.json; p3
7	N2_migkas_cov	**NEW**	Migkas digitised; p1
8	d2_web	regenerated (bodies replace PS web)	p5_c_p8
9	d3_stoch	regenerated	p6_p8
10	N5_seed_distribution	**NEW**	p8_field_consistent
11	N3_joint_summary	**NEW**	p2, p5 run a ($f = 1$), p5_c_p8, p1, p6_p8
Table 5	N4_tt_stats (no PNG)	**NEW**	p6_p8

Not carried over from rev6 (no script, or not regenerable from the current solve): Fig. 1 top (preprint sketch); sky-orientation sequence; MAX/MIN bands on Migkas Fig. 8; ring harmonics; web slab and kernel ladder (PASS labels); target matches (two figures); stage-A fit; $m(r)$ at 42 Gpc; Migkas direct fits (poly variants); passing-universe domains; combo10 low multipoles (hand-placed $\ell = 220$ hump); $R = 4, 5, 6$ ladder; degeneracy slices; damped-harmonic residuals; harmonic decomposition; anomaly maps and spot scores; Migkas+rotation (needs the spreadsheet; moved to `fig7_stale_noscript/`).

C Changes from rev6

1. One background (LTB famP) and one $D = 34.96\,\mathrm{Gpc}$ everywhere; H_0, age and R_{ls} derived (§2.2); tensions stated. Eq. (2) of rev6 ($t_0 = 13.8\,\mathrm{Gyr}$ FRW) removed.
2. D chosen by a Migkas profile (NEW Fig. 2).
3. Lock laws re-solved at $D = 34.96$ (Table 3); $\delta = 0.031$ vs $2\mathcal{D}$ stated.
4. Cluster covariance, n_{eff}, dipole and dipole + quadrupole comparison, BIC (NEW §5.2); significance described as qualitative.
5. Deterministic CMB sky recomputed at R_{ls} (§5.3); compensating bodies on the consistent background with a true-error metric (§5.4); Migkas fit checked unchanged.
6. CMB kernels moved from $D = 42$, $R = 4$ to the rev7 values; 15-parameter P_δ (running, amplitude tilt, damping exponent); TT table rebuilt with $\Delta\chi^2$, ΔBIC, fit objective and one peak finder.
7. 1,000-realisation distribution for CoE and ΛCDM (NEW §5.7).
8. ΛCDM/CDM web replaced by the solved bodies (Fig. 8).
9. Anomaly score withdrawn; harmonic decomposition, sky orientations and rings not carried over.
10. Single-column layout; all figures regenerated or new with scripts beside them; rev6 figures without scripts dropped (Appendix B).
11. Foreman et al. (2010) added; Eisenstein & Hu, Press & Schechter, Sheth & van de Weygaert dropped with the web.
12. Appendix D (referee response) added.

D Response to referee, audit and internal review

Sources: referee report (M1–M9, minor 1–16), mathematics-and-code audit (A1, E1–E12, B1–B12), internal five-pass review (IR1–IR10 = its top-10 issues; IRm = its minor list).

Item	Issue	Response in rev7
M1	71-bin χ^2 not valid	Overlap covariance, pseudo-inverse, $n_{\mathrm{eff}} = 6.0$ (§5.2); significance qualitative. Bootstrap not possible: catalogue not on machine.
M2	no harmonic comparison	Dipole ($k = 3$) and dipole + quadrupole ($k = 5$) under the same likelihood; CoE lowest χ^2_C and BIC, not significant (Table 4).
M3	lock relates wrong quantities	Not redone. Assumptions listed, plus new point that $d \gg R_{\mathrm{ls}}$ (§5.1). Open.
M4	background inconsistent	Resolved: one LTB background, one D (§2.2); new tensions (H_0, age) stated.
M5	CMB input vs output	Fit objective vs reported χ^2, $\Delta\chi^2/\Delta$BIC, acoustic terms called inputs (§5.5, Table 5).
M6	11/12 anomaly score	Withdrawn (§5.10); null not computed.
M7	residual-driven building	Parameter table (App. A), BIC throughout. No holdout yet; listed as next step 5.
M8	moving-lens frame; quadrupolar modulation	Not resolved. Quantified scale of the relative velocity on rev7 background ($\approx 0.02c$ vs $4c$) (§5.9). Open, priority 1.

Item	Issue	Response in rev7
M9	pass rates	1,000-sky distribution for current field and ΛCDM; 26 % pass strict, 99.3 % loose cut (§5.7). rev6 web ensemble not re-run.
minor 1	abstract mixes fitted/predicted	Abstract rewritten with predicted/fitted separated, significances given.
minor 2	d called Migkas depth	Corrected (Table 2, §5.1).
minor 3	R^2 with fitted alignment	Dipole-rotation figure not carried over; B not claimed.
minor 4	"10^{-15}" root-finding claim	Removed.
minor 5	$\delta = 0.031$ vs $2\mathcal{D}$	Stated with both L values (Eq. 7).
minor 6	tilt/roll "no d.o.f."	Orientation sequence not carried over.
minor 7	Eq. 19 example	Superseded by the new P_δ; old example removed.
minor 8	mixed units	Gpc and Gpc^{-1} throughout §5.5.
minor 9	peak lists differ	One finder for all three spectra (Table 5).
minor 10	two χ^2 for A	Harmonic decomposition dropped; one 71-bin χ^2 convention.
minor 11	B r^2 falls	Dropped with the decomposition.
minor 12	$\ell = 50$ step from log	Superseded: c_ℓ refresh correlation reported (0.899, 0.978).
minor 13	PASS label	Figure dropped; pass rate recomputed (§5.7).
minor 14	preprint figures	Dropped (App. B).
minor 15	two orientations	One axis $l_0 = 19.87°$; orientation figures dropped.
minor 16	references	Completed; Foreman et al. added.
A1	verified items	Kept where still used (volumes now on rev7 background).
E1	Eq. 19 example	As minor 7.
E2	one background	As M4.
E3	horizon $< d$	On rev7, $R_{\mathrm{ls}} = 0.19\,\mathrm{Gpc} \ll d$; stated as a lock limitation.
E4	δ rounding	As minor 5.
E5	withdrawn	–
E6	$D = 42$ vs 58.2	One D; kernel factor 42 replaced by 34.96.
E7	Fig. 1 label	Removed.
E8	axis longitudes	One l_0.
E9	conditional contours	Degeneracy figure dropped; no uncertainties claimed (§5.5).
E10	peak finders	As minor 9.
E11	A χ^2 25.54 vs 45	As minor 10.
E12	stated vs actual objective	Objective and optimiser stated (§5.5).
B1	bulk velocity	As M8.
B2	quadrupolar modulation	As M8; open.
B3	A_F is a data fit	Score using it withdrawn.
B4	fitted dipole rotation	Figure dropped; B not claimed.
B5	hand-placed $\ell = 220$ hump	combo10 figure dropped; no hand-placed terms in rev7 figures.
B6	minimised ≠ reported χ^2	Both in Table 5.
B7	PASS label	As minor 13.
B8	Nelder–Mead, no uncertainties	12 multistarts; uncertainties still [TO COMPUTE].
B9	single seed	1,000 seeds (§5.7).
B10	tangent slope vs lock	$m(r)$ at 42 Gpc dropped; one D.
B11	reproducibility	Every figure has its script (App. B).

Item	Issue	Response in rev7
B12	things done well	Kept: candour, worked derivations, validity notes.
IR1	residual CMB multipole; off-centre bounds	Recomputed at $R_{\rm ls}$ (2.3×10^{-5}), cancelled to 1.5×10^{-6}; bounds discussed (§6).
IR2	correlated cones	As M1.
IR3	inconsistent background	As M4.
IR4	Fourier beats CoE	As M2 (CoE now lowest BIC; not significant).
IR5	11/12 post hoc	As M6.
IR6	acoustic peaks inserted	Stated as inputs; $\Delta\chi^2 = +58$, $\Delta\mathrm{BIC} = +128$; blind test pending.
IR7	lensing/shear; 48.9 vs 149.9	Shear compensation now the reported, reproducible bodies run (p5_c_p8); CMB lensing not added.
IR8	CatWISE kernel	As M3.
IR9	mean vs seed-42	Both reported, plus distribution.
IR10	SN acceleration; literature	Literature added; $D_L(z,\hat{n})$ not computed (next step 7).
IRm	minor list	2.06e-9, peak lists, $D = 42$ vs 58.2, PASS, δ rounding, references: handled above. s_w on bound: rev7 $s_w = 0.01096\,\mathrm{Gpc} = 0.23s$, inside its allowed range $(0.05s, 2.5s)$, $s = R/4$. f, $p(s)$, K_G: $f = 0.26$ defined; $p(s)$, K_G still unverified.